

QA Report

Research Title:	Multi-spectral Change Analysis in Sentinel-2 Satellite Imagery using Difference Subspace & U-Net
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Question 1

Why do DS and PCA-diff sometimes highlight regions that are not labeled as "change" in OSCD?

Answer at the time of the presentation

Because OSCD labels only semantic change, while DS/PCA-diff respond to spectral change. Vegetation shifts, illumination differences, or atmospheric effects produce real spectral differences even if OSCD doesn't mark them as change. That's why these priors are useful, they reveal information the labels don't capture.

Improved answer after the presentation:

OSCD change masks are defined for specific land-cover / land-use transitions (e.g. construction, urbanization, deforestation) and are not intended to flag every radiometric difference between the pre- and post-event Sentinel-2 images. DS and PCA-diff, on the other hand, are purely spectral and unsupervised: they build subspaces or principal components from all valid spectral vectors and then measure how strongly each pixel pair projects onto the "difference" directions. As a result, any systematic spectral shift -seasonal vegetation change, soil moisture, haze, thin clouds, snow, or illumination changes- is highlighted, even when the OSCD mask does not label it as change. In Phase 2 we also see this mismatch: adding DS/PCA-diff sometimes improves AUROC (better ranking of pixels) but can slightly reduce IoU/F1 against OSCD, because the network becomes more sensitive to these extra spectral changes. This behavior is still valuable for disaster damage mapping, where labels are often incomplete: DS/PCA-diff are best interpreted as spectral-change priors and explanatory overlays, not as exact predictors of the OSCD semantic masks.[2]

Question 2

In phase 1 datasets. you mentioned that you use the MultiSenGE dataset to test out Difference Subspace and if it produces interpretable change maps. Regarding the two images you use to extract the difference or change from, how do you choose them exactly from the dataset?

Answer at the time of the presentation

For a certain geographical patch in the dataset, I just use the earliest timestamped image, and the latest timestamped image available, there are a lot of images for a certain patch taken at different times not just two images. reason why I chose the earliest Vs. latest approach is to detect change, there needs to be change in the first place, so I just choose earliest Vs latest two images

for a patch that are available in the dataset, sometimes the time between these two timestamps is months, week, or days, the period between satellite images taken and sorted in the dataset isn't equal. I'm actually planning to implement a method to detect the change for the multiple images in between the first taken and last taken image instead of only comparing the earliest Vs. latest images for a certain patch, that could allow me to extract granular change and potentially use a method called Second-order Difference subspace, a method built on the first-order difference subspace that I used in my project, to understand the progression of change for a certain region, you can imagine Second-order difference subspace corresponds to the velocity and acceleration of a moving object

Improved answer after the presentation

In Phase 1, MultiSenGE is used as an unlabeled testbed to visualize how DS behaves on realistic Sentinel-2 time series. For each spatial patch, we group all acquisitions, sort them by date, filter out obviously invalid images, and then form a bi-temporal pair using the earliest and latest valid dates. This gives the maximum temporal baseline and makes it likely that any major land cover evolution appears in the DS map, but it also means that the time gap is highly variable and seasonal effects (for example winter vs. summer vegetation or snow) can dominate the spectral differences. As a follow up, the planned direction is to move from a single earliest-latest pair to controlled multi-temporal pairings (or sequences), e.g. comparing within-season pre/post windows, aggregating several dates around each event, and then exploring first- and second-order DS to capture both level and progression of change. This should reduce purely seasonal artifacts and make the DS maps more directly interpretable for damage-related scenarios.

Comments

Comment 1

if your method can overcome seasonal change (such as snow), it will be great.

Investigation after the presentation

Seasonal effects like snow are a known difficulty for spectral change detection, snow strongly alters reflectance in visible and NIR bands and will inevitably produce high DS / PCA-diff scores, even without semantic damage. This is consistent with reports that Sentinel-2 change-detection methods can overestimate land conversion when true change is rare and radiometric or seasonal variability dominates the signal [3]. In our pipeline, the most direct countermeasures are: (1) season aware pairing of pre/post images (e.g. only comparing images from similar seasons for a given area), (2) explicit snow and cloud masking before computing DS/PCA-diff, for example via NDSI and standard Sentinel-2 cloud masks, and (3) in Phase 3, focusing DS computation on building or built-up regions rather than entire tiles, so that the priors are driven more by structural changes than by background snow cover. At the current stage DS/PCA-diff are not season-invariant, but the comment shapes the next step for me which is making season handling and snow masking part of the standard preprocessing for any damage oriented application. really good comment.

Comment 2

IR-MADは一般に強力なマルチバンド手法ですが、なぜ本研究では弱かったと解釈していますか？ Sentinel-2のスペクトル特性の影響でしょうか？

Investigation after the presentation

IR-MAD is usually a strong multi-band change detector, it uses canonical correlation analysis and an iteratively reweighted scheme to build a stable "no-change" background and then emphasizes deviations from it [1]. In our OSCD experiments however, IR-MAD was weaker than DS and PCA-diff. My current interpretation is that several factors interact: (1) OSCD tiles are relatively small and often dominated by urban cores or engineered changes, so the fraction of truly stable pixels may be too low for IR-MAD to fully benefit from its reweighting. (2) we fed all 13 Sentinel-2 bands without careful band selection or regularization, which may make canonical correlation more sensitive to noisy or highly atmospheric bands. (3) the evaluation metric is semantic OSCD change masks, not generic spectral change, so some of the strong spectral differences detected by IR-MAD are counted as false positives. In contrast, DS and PCA-diff were slightly better tuned to our normalization and thresholding pipeline. A more fair comparison would involve explicit band selection, robust covariance estimation, and calibrated thresholds for IR-MAD on Sentinel-2, but within the scope of this project it served as a conservative baseline and highlighted that DS can be competitive even compared to this well established method.

Reflection on My Presentation

The questions about DS/PCA-diff highlighting unlabeled regions showed me that I need to explain the difference between spectral and semantic change earlier and more clearly. In the talk, this nuance only appeared in the later discussion, so some of the qualitative figures may have looked like obvious "errors" rather than label mismatches. The MultiSenGE question made me realize that pairing strategy is not just a technical detail, choosing earliest-latest is simple but has strong implications for seasonality and interpretability, and I should emphasize this in both slides and narration. The comments on seasonal change and IR-MAD emphasize that strong baselines require careful adaptation to each dataset, I should communicate more transparently which methods were tuned and which were used as rough references. this helped me see where to tighten the research details and technical assumptions.

References

- [1] A. A. Nielsen, "The Regularized Iteratively Reweighted MAD Method for Change Detection in Multi- and Hyperspectral Data," *IEEE Transactions on Image Processing*, vol. 16, no. 2, 2007. doi:10.1109/TIP.2006.888195
- [2] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully Convolutional Siamese Networks for Change Detection," in *Proceedings of the 2018 25th IEEE International Conference on Image Processing (ICIP)*, pp. 4063–4067, 2018. doi:10.1109/ICIP.2018.8451652
- [3] O. Close, S. Petit, B. Beaumont, and E. Hallot, "Evaluating the Potentiality of Sentinel-2 for Change Detection Analysis Associated to LULUCF in Wallonia, Belgium," *Land*, vol. 10, no. 1, article 55, 2021. doi:10.3390/land10010055